

Subsurface imaging in shield tunnelling and pipeline localisation via a robust cosmic-ray muography system: a field study

Z. Yao¹, Y. Zhou^{2,*}, L. Huang^{3,*}, S. Liu¹, Y. Wang¹, Z. Zhang¹, Y. Wang², H. Fei², B. Yu⁴, J. Lei³, T. Song³, Z. Zhao³, Z. Xiong³, K. Li³, & Z. Zhang⁵

¹University of Science and Technology of China, Hefei, China

²Jianwei Scientific Instruments (Anhui) Technology Co., Ltd., Hefei, China

³Shenzhen Metro Group Co., Ltd., Shenzhen, China

⁴Jianwei Geophysical Exploration (Shenzhen) Technology Co., Ltd., Shenzhen, China

⁵China Railway No.8 Engineering Group Co., Ltd., Chengdu, China

*Corresponding authors: Yong Zhou (zhouyong@jwky.com);

Liping Huang (huanglp@shenzhenmc.com).

ABSTRACT: Uncertain ground conditions and buried utilities pose challenges to shield tunnelling. Cosmic-ray muography uses the attenuation of naturally occurring penetrating particles to investigate subsurface density variations, complementing conventional geological surveys without an artificial radiation source. However, mounting a muon detector on an operating tunnel boring machine exposes it to persistent mechanical vibration, elevated temperatures and high humidity. These conditions can compromise detector stability and data quality. This paper presents a robust three-layer Micromegas muography system developed for these conditions and two field applications on Shenzhen Metro Line 22. Active temperature and humidity control, combined with pneumatic vibration isolation, enabled stable muon data acquisition in the shield construction environment. In the first application, the system was deployed on operating tunnel boring machines, and the collected measurements were used for three-dimensional reconstruction of the surrounding ground density. The reconstruction incorporated the tunnel geometry and used borehole stratigraphy to represent the geological background. Comparison with the borehole-derived geological model suggested a possible low-density region above and to one side of the tunnel. In the second application, observations from four detector positions were used to investigate a suspected buried pipeline. Directional features extracted using U-Net models were combined across multiple viewpoints to estimate a candidate pipeline axis in three-dimensional space. The inferred location informed recommendations provided to the construction team for pipeline avoidance during shield advance. Together, these field applications demonstrate the technical feasibility of the developed instrumentation and the practical value of muographic data for subsurface engineering.

Keywords: muography; shield tunnelling; density reconstruction; pipeline localisation

1 INTRODUCTION

Uncertain ground conditions and buried utilities can interrupt shield advance and threaten surrounding infrastructure. Boreholes provide direct but spatially discrete geological information, while urban access restrictions and ongoing construction can constrain surface investigations. Measurements acquired from within a tunnel can complement these surveys by providing information about the ground surrounding the excavation. Muon detection systems designed for deployment within tunnels offer a way to obtain these observations during shield construction.

Cosmic-ray muography exploits muons produced when primary cosmic rays interact with the atmosphere. Their penetrating ability allows a fraction of them to reach underground detectors after traversing soil and rock. Muon imaging can use either attenuation or scattering; the transmission approach used here measures the directional flux remaining after passage through the target (Procureur 2018). A tracking detector reconstructs particle directions from hit positions measured in multiple sensitive layers. The measured muon flux is compared with an open-sky reference and interpreted using a muon spectrum and an energy-loss model to estimate opacity. This quantity represents the density integrated along the muon path. Measurements from different directions and positions can then be combined with tunnel geometry and geological information to reconstruct

density variations. This provides a passive observation method without an artificial radiation source or direct access to the target.

Previous studies have used underground muon measurements to investigate overburden density and concealed structures around tunnels. Four detector locations beneath the Los Alamos mesa were used to reconstruct the overburden density and rock–air interface (Guardincerri et al. 2017). A railway-tunnel survey identified density anomalies associated with hidden construction shafts and an overburden void subsequently confirmed by the railway authorities (Thompson et al. 2020). Measurements along a subway station and tunnel detected changes in overburden structure and anomalies associated with construction features (Mao et al. 2023).

Applying muography on an operating tunnel boring machine (TBM) requires reliable data acquisition under construction conditions. Mechanical vibration, elevated temperatures and humidity can affect detector stability, electronics and data quality. Confined installation space and access for maintenance also influence instrument design. Stable acquisition must therefore be supported by environmental control and mechanical isolation. As the shield advances, the detector also changes position relative to the ground, requiring the measurements to be analysed with their corresponding observation geometry.

This paper presents a robust three-layer Micromegas muography system and two field applications on Shenzhen Metro Line 22. The instrument combines active temperature and humidity control with pneumatic vibration isolation for TBM-mounted measurement. The first application uses underground muon data, tunnel geometry and borehole stratigraphy to reconstruct geological density variations. The second combines U-Net-based extraction of directional features with multi-view geometry to estimate a buried pipeline axis. Together, the applications examine both sustained field operation and the use of muography for subsurface investigation during shield construction.

2 MUOGRAPHY SYSTEM AND FIELD OPERATION

The system uses three Micromegas tracking layers fabricated by thermal bonding (Feng et al. 2021). Each layer has a $40 \times 40 \text{ cm}^2$ sensitive area; adjacent layers are approximately 15 cm apart. The detectors use argon with a small fraction of carbon dioxide. Muon ionisation releases electrons that undergo avalanche multiplication. The amplified charge induces signals on orthogonal strips with a 0.4 mm pitch. To improve the signal-to-noise ratio, four adjacent strips are combined into a single readout channel, giving an effective readout pitch of 1.6 mm. The two orthogonal strip sets provide two-dimensional hit positions. Fitting these positions across the three layers reconstructs each muon trajectory. Compact front-end cards use ADAS1128 current-to-digital converters to integrate and digitise the signals (Wang et al. 2025; Analog Devices n.d.). A separate data acquisition (DAQ) board distributes clock and trigger signals, applies a coincidence window to front-end triggers and aggregates events. Common trigger identifiers associate signals across layers (Liu et al. 2025). Optical fibres link the front-end cards to the DAQ board, which transfers data to the computer via USB 3.0.

The main enclosure houses the detectors, readout electronics, power supplies and temperature and humidity controls; a separate enclosure houses the gas supply. Figure 1 shows the enclosure design with dimensions (a), internal structure (b) and TBM deployment (c). Four pneumatic isolators separate the inclined frame from the shield support platform, as shown in Figure 1(c).

The system operated at depths of approximately 15–30 m in the Guihua Station–Songyuanxia Station interval from 6 November 2025 to 27 July 2026, covering 1371 rings (428 left-line and 943 right-line). Installation distances behind the cutterhead were about 5 m on the left line and 15 m on the right. Monitored temperatures remained below 40 °C and relative humidity below 70%. Normal acquisition totalled 2965.5 h on the right line: 85.2% of the 3480.0 h right-line monitoring record, or 97.2% after excluding external power and network interruptions from that duration. Downtime comprised 84.7 h of maintenance, 401.5 h of power loss and 28.3 h of network interruption.

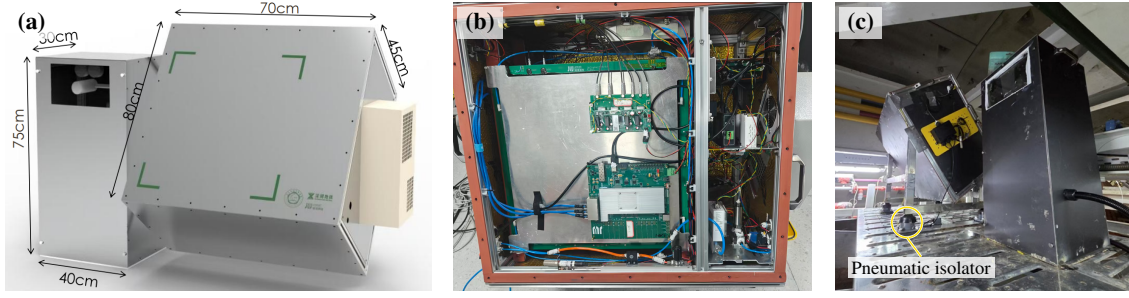


Figure 1. Muography system: (a) enclosure and exterior design with dimensions; (b) photograph showing the internal structure; (c) deployment on the tunnel boring machine.

3 TBM-MOUNTED GEOLOGICAL DENSITY IMAGING

3.1 Measurements and geological model

Muon observations at successive shield positions were combined with borehole stratigraphy to investigate density variations around the tunnel near the Fuzhou Avenue bridge. The primary dataset contains 5240 and 6840 selected underground tracks at rings 1672 and 1679, respectively, and 457,255 matched open-sky tracks. Underground exposures were 21.37 and 34.00 h; the two sky exposures were 20.97 and 66.52 h. The same detector hardware, track-reconstruction procedure and selection criteria were used for the open-sky and underground measurements.

In the geological survey frame, Y is upward, Z points towards decreasing ring number and X is transverse. Ring 1672 defines the longitudinal origin and vertical track reference. The detector centres at rings 1672 and 1679 are represented by $(2, 0, 0)$ and $(2, 0.27, -12.66)$ m, respectively. The initial detector-normal elevation was 33.2° , varying within $\pm 0.6^\circ$ during advance.

The geological model uses 11 borehole control points along the tunnel. Table 1 summarises the stratigraphy and assigned densities at five boreholes around the measurement interval. Each layer is assigned the reference density specified in the geotechnical parameter table, applied consistently across boreholes. Layer boundaries are interpolated along the tunnel and extended uniformly across X . The construction drawings specify tunnel inner and outer diameters of 8.0 and 8.8 m. The excavated cavity and 0.4 m lining are modelled separately, with a lining density of 2.4 g cm^{-3} .

Table 1. Borehole locations and layer-top depths below ground (m), with assigned densities (g cm^{-3}).

Density	TJL-39 (-0.87, 62.05)	TJL-38 (8.32, 17.19)	TJL-37 (10.02, -17.19)	THS-2-1 (8.86, -43.57)	THS-2 (1.38, -73.94)
1.95	0.00	0.00	0.00	0.00	0.00
2.20	3.00	0.00	5.00	6.60	6.00
1.85	3.00	8.50	5.00	6.60	6.00
1.95	20.00	18.90	18.50	18.00	17.60
2.20	22.50	25.40	24.80	26.00	20.00
2.25	23.70	34.30	42.30	31.00	22.00
2.40	23.70	34.30	44.50	31.00	22.90
2.45	26.50	37.80	—	33.80	27.40

Coordinates in parentheses are (X, Z) in metres. All borehole identifiers have the prefix M33Z3-. Layers are ordered from shallow to deep; equal successive depths indicate zero thickness.

3.2 Joint density reconstruction

The reconstruction estimates density corrections to the borehole reference model on a 2 m cubic grid, with $\rho_j = \rho_{0,j} + \delta_j$, where $\rho_{0,j}$ is the reference density and δ_j its correction in voxel j . For

ray q from measurement position r , the opacity is

$$X_{rq} = X_{0,rq} + \sum_j A_{rqj} \delta_j, \quad (1)$$

where $X_{0,rq}$ is the reference opacity and A_{rqj} is the rock path length through voxel j , calculated from ray–voxel intersections (Siddon 1985). The tunnel interior is treated as air, and the lining contribution remains fixed. Opacities are converted to muon transmission using the spectrum of Guan et al. (2015) and standard-rock energy-loss ranges (Groom et al. 2001).

The measured sky and underground counts, m_b and n_{rb} , are treated as independent Poisson observations, giving the joint likelihood

$$L(\delta, \lambda) = \prod_b \frac{\mu_{s,b}^{m_b} e^{-\mu_{s,b}}}{m_b!} \prod_{r,b} \frac{\mu_{rb}^{n_{rb}} e^{-\mu_{rb}}}{n_{rb}!}, \quad \mu_{s,b} = t_s \lambda_b, \quad \mu_{rb} = \eta t_r \lambda_b P_{rb}(\delta), \quad (2)$$

where b indexes angular bins and r the two underground positions. The expected counts $\mu_{s,b}$ and μ_{rb} depend on the sky and underground exposures t_s and t_r , open-sky bin rate λ_b , and bin-averaged transmission P_{rb} . The vectors δ and λ collect the density corrections and bin rates. A common factor $\eta = 0.53$ normalises both positions, estimated from observed-to-predicted count ratios in background directions excluding known bridge-pile locations in the construction drawings.

The estimates minimise the negative log-likelihood together with geological and spatial penalties:

$$(\hat{\delta}, \hat{\lambda}) = \arg \min_{\delta, \lambda} \left[-\log L(\delta, \lambda) + \frac{1}{2} \sum_j w_j \delta_j^2 + \frac{1}{2} \sum_{(j,k) \in \mathcal{N}} v_{jk} (\delta_j - \delta_k)^2 \right], \quad (3)$$

where hats denote fitted estimates and $\mathcal{N}$ contains each face-sharing voxel pair once. The first term fits the counts; the second limits departures from the borehole model with prior weights w_j ; the third limits abrupt differences between neighbouring corrections with weights v_{jk} . Prior weights decrease with distance from boreholes, and neighbour weights are reduced across geological layer boundaries.

For each trial density distribution, the open-sky rates are estimated analytically by maximising L . Density corrections are optimised using bounded L-BFGS-B (Zhu et al. 1997). The reconstruction domain is $X = [-48, 30]$, $Y = [-6, 36]$ and $Z = [-24, 42]$ m. Counts are fitted in 151 adaptive angular bins with widths of 6° or 12° . Model transmission within each bin is weighted by the open-sky angular distribution.

3.3 Reconstructed density distribution

Figure 2 compares the count-derived effective opacity with the borehole reference at each measurement position. Both positions show negative opacity differences at large zenith angles, while positive differences are more prominent at ring 1679. Negative values indicate less integrated density along the viewing direction than predicted by the geological model.

Combining the two viewing geometries gives the density-correction distribution in Figure 3. A possible low-density region appears above and to the negative- X side of the tunnel. A positive correction near ring 1679 reaches $+0.57 \text{ g cm}^{-3}$ at $(1, 9, -9)$ m, a location directly sampled by that position.

Regional means are evaluated over a fixed set of voxels above the tunnel on its negative- X side, selected from a preliminary joint reconstruction and used unchanged for all three fits. The joint mean correction is -0.36 g cm^{-3} ; separate fits to rings 1672 and 1679 give -0.21 and -0.24 g cm^{-3} , respectively. Both separate fits therefore give a negative regional mean, although their spatial distributions differ.

Cross-position predictions provide a further consistency check using Poisson deviance, where smaller values indicate closer agreement. Predicting ring 1679 from the ring-1672 reconstruction reduces the deviance from 801.4 to 770.6; the reverse prediction reduces it from 272.1 to 256.9. These improvements are consistent with the negative regional means.

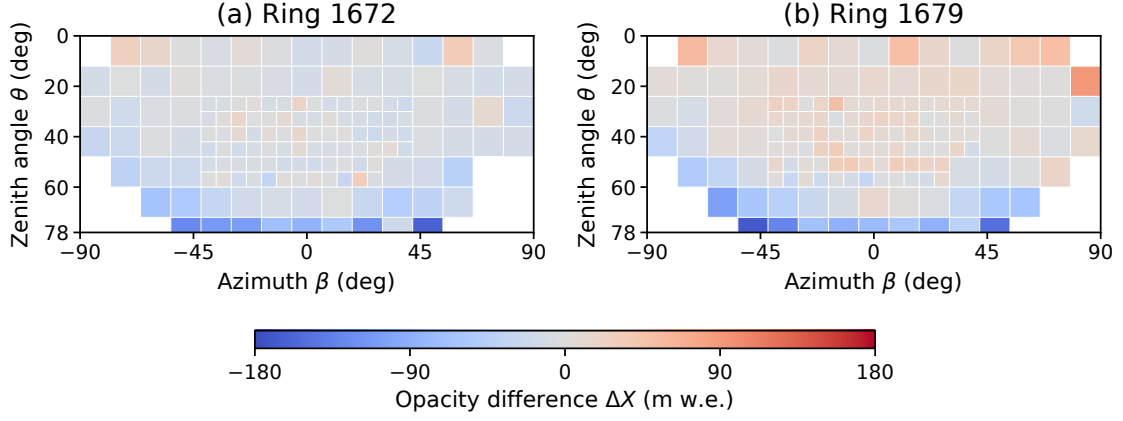


Figure 2. Effective opacity differences, $\Delta X = X_{\text{obs}} - X_{\text{reference}}$, for (a) ring 1672 and (b) ring 1679.

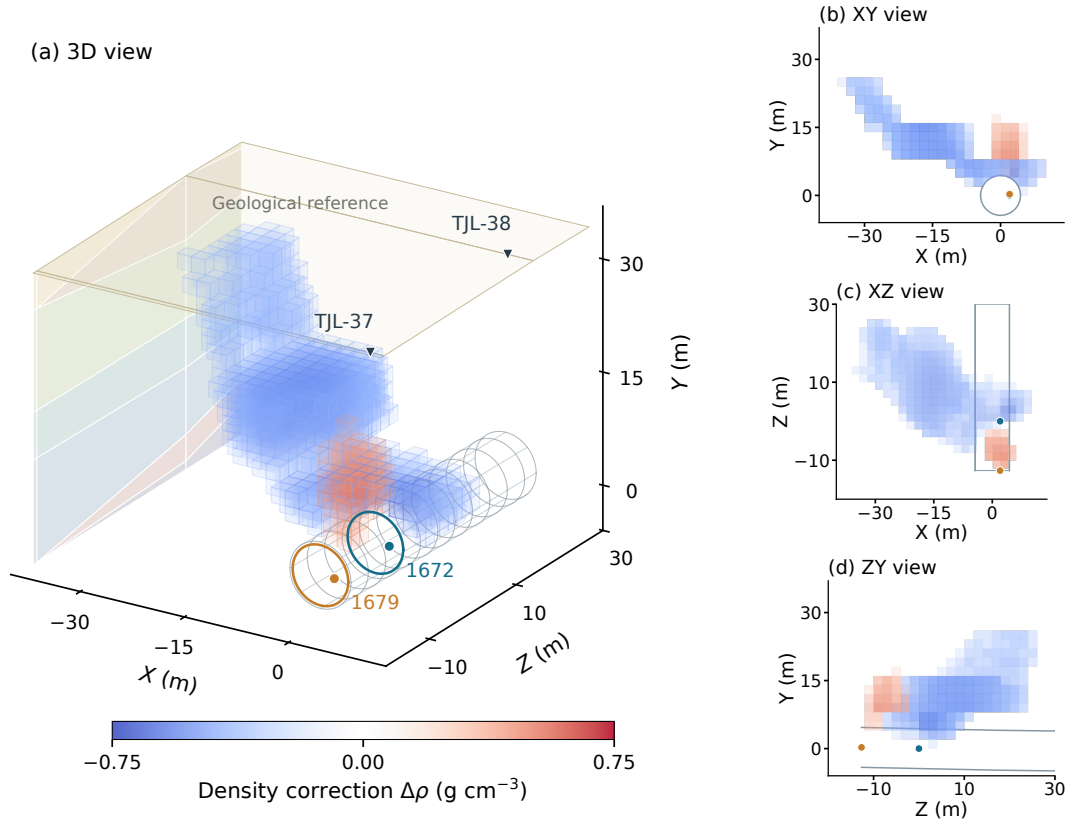


Figure 3. Joint density corrections relative to the borehole model: (a) three-dimensional rendering and (b–d) XY, XZ and ZY views. Only 2 m voxels with direct ray coverage and $|\Delta\rho| \geq 0.35 \text{ g cm}^{-3}$ are shown; transparency decreases with $|\Delta\rho|$.

4 FOUR-POSITION PIPELINE SEARCH

4.1 Survey geometry and directional features

Conventional detection methods had not located the pipeline at the position indicated in the original drawings, leaving uncertainty about its location relative to the planned tunnel alignment. The muography survey was therefore undertaken to investigate whether additional positional information could be obtained to inform pipeline-avoidance measures during shield advance.

The detector was deployed sequentially at four positions within the tunnel, as shown in Figure 4. Acquisition times at P0–P3 were 22.8, 21.7, 32.5 and 22.0 h, respectively. Tracks were binned into

36×38 directional maps, with 21,084–31,309 underground counts per position and 78,718 counts in the shared open-sky map.

The coordinate origin for this survey is the centre of ring 252; X is along the tunnel opposite to advance, Y points horizontally towards the reference wall and Z is upward. Table 2 gives the instrument centres and orientations used in the reconstruction. Pitch is the elevation of the detector normal; $\alpha = 0$ faces positive Y , with positive rotation towards negative X . The geometric model follows the construction drawings, with tunnel inner/outer radii of 2.75/3.10 m and ground at $Z = 16.8$ m.

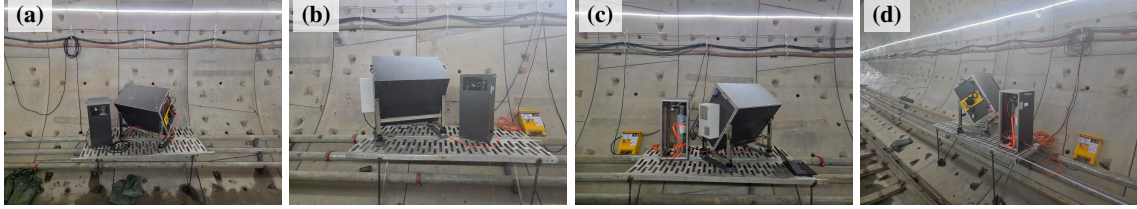


Figure 4. Field deployment for the pipeline survey at (a) P0, (b) P1, (c) P2 and (d) P3.

Table 2. Pipeline detector centres and orientations in the ring-252 frame.

Position	X (m)	Y (m)	Z (m)	Pitch ($^\circ$)	α ($^\circ$)
P0	−0.05	1.38	−1.50	36.3	−32.4
P1	3.75	1.38	−1.50	34.3	17.1
P2	7.50	1.48	−1.50	34.9	35.0
P3	−3.80	1.60	−1.50	35.8	−77.0

A position-specific U-Net (Ronneberger et al. 2015) is trained on simulated directional images covering different pipeline orientations and including cases without a pipeline. The trained network is then applied to the corresponding field observations with its parameters fixed. The network outputs are relative scores, not calibrated probabilities of pipeline presence. Evaluation on 50,000 simulated test samples per position gave pixel ROC-AUC values of 0.9585–0.9703. As shown in Figure 5, all four field score maps contained an extended ridge.

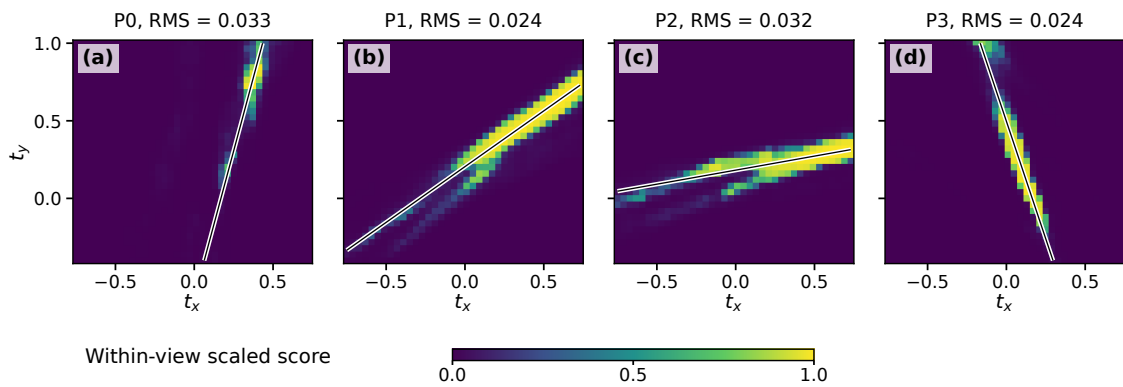


Figure 5. Pipeline directional maps at (a) P0, (b) P1, (c) P2 and (d) P3. Colours represent within-view scaled network scores; lines mark fitted ridges. Coordinates are detector-local slopes; RMS denotes the weighted root-mean-square residual transverse to each ridge.

4.2 Three-dimensional reconstruction and engineering use

Weighted principal-component analysis fits a straight ridge to high-score pixels in each directional map, using detector-local slopes $t_x = x_{\text{local}}/z_{\text{local}}$ and $t_y = y_{\text{local}}/z_{\text{local}}$. The detector position and orientation transform each ridge into a global backprojection plane:

$$\mathbf{n}_i^T(\mathbf{x} - \mathbf{c}_i) = 0, \quad (4)$$

where $\mathbf{c}_i$ is the detector centre and $\mathbf{n}_i$ is the unit plane normal. The candidate pipeline direction is found by minimising

$$\hat{\mathbf{d}} = \arg \min_{\|\mathbf{d}\|=1} \left[\sum_i w_i (\mathbf{n}_i^T \mathbf{d})^2 + \gamma d_z^2 \right], \quad (5)$$

where the empirical weights w_i are proportional to ridge line-likeness divided by transverse residual and satisfy $\sum_i w_i = 1$. Ridges with stronger linearity and less transverse scatter receive greater weight. The first term favours a direction parallel to the four planes; the second penalises the vertical component d_z , with $\gamma = 1$ imposing a soft preference for a near-horizontal pipe. A reference point is obtained by minimising the weighted sum of squared distances to the planes.

The fitted candidate axis is

$$\mathbf{x}(\ell) \simeq [-3.41, 0.38, 6.50]^T \text{ m} + \ell[0.69, 0.73, -0.05]^T, \quad (6)$$

where ℓ is distance along the axis in metres. As shown in Figure 6, the candidate passes above the tunnel near ring 252.

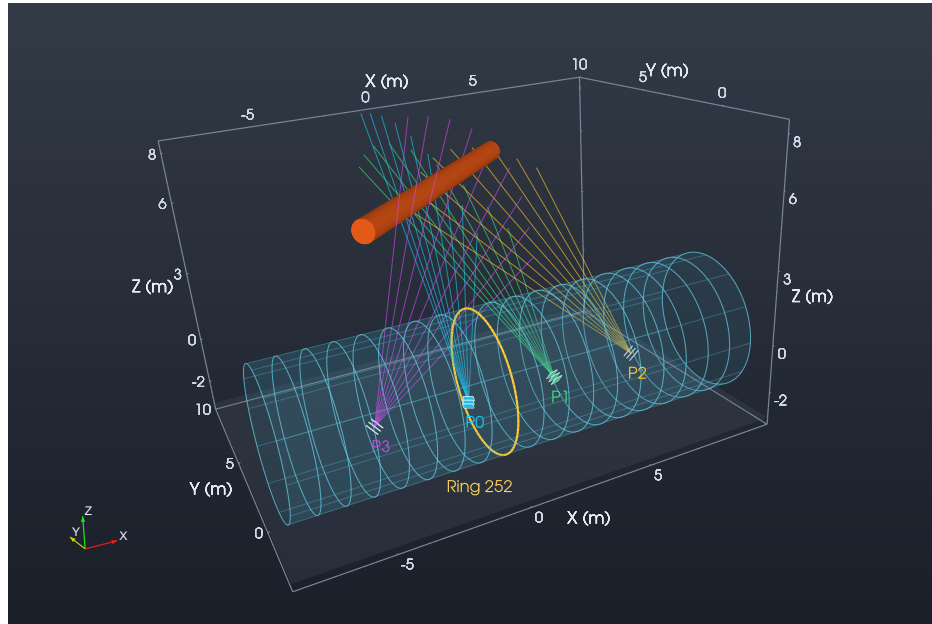


Figure 6. Three-dimensional pipeline candidate axis with the horizontal soft constraint. Coloured ray fans represent the backprojection planes from P0–P3; the orange tube shows the candidate pipe with its reported diameter of 0.8 m.

During the field campaign, the muography team provided candidate pipeline positions and shield-avoidance recommendations to the construction team, which reported implementing avoidance measures based on this information.

5 CONCLUSIONS

The field instrument enabled muon acquisition in an operating shield-tunnelling environment through environmental control and pneumatic isolation. Joint reconstruction from two measurement positions indicated a possible low-density region above the tunnel relative to the borehole reference, with a mean density correction of -0.36 g cm^{-3} within the selected comparison region. Cross-position predictions supported this interpretation. The four-position pipeline survey yielded a candidate three-dimensional axis and provided positional information for shield-avoidance planning. These applications demonstrate the potential of transportable muography to support geological anomaly screening and buried-pipeline assessment during tunnel construction. Further work should improve quantitative localisation through matched reference measurements and additional viewpoints.

ACKNOWLEDGEMENTS

The authors acknowledge Shenzhen Metro Group Co., Ltd. for providing access to the field sites and coordinating the measurements on Shenzhen Metro Line 22, the University of Science and Technology of China for research support, and the Jianwei teams for providing detector equipment, technical support and maintenance throughout the field campaigns. The authors also acknowledge the assistance of the site personnel with detector installation and data acquisition.

REFERENCES

- Analog Devices n.d. *ADAS1128 Datasheet and Product Info*. URL: <https://www.analog.com/en/products/adas1128.html> (visited on 09/29/2026)
- Feng, J., Zhang, Z., Liu, J., Qi, B., Wang, A., Shao, M. & Zhou, Y. 2021. A thermal bonding method for manufacturing Micromegas detectors. *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment* 989: 164958.
- Groom, D. E., Mokhov, N. V. & Striganov, S. I. 2001. Muon stopping power and range tables: 10 MeV–100 TeV. *Atomic Data and Nuclear Data Tables* 78(2): 183–356.
- Guan, M., Chu, M.-C., Cao, J., Luk, K.-B. & Yang, C. 2015. A parametrization of the cosmic-ray muon flux at sea-level. arXiv:1509.06176.
- Guardincerri, E., Rowe, C., Schultz-Fellenz, E., Roy, M., George, N., Morris, C., Bacon, J., Durham, M., Morley, D., Plaud-Ramos, K., Poulson, D., Baker, D., Bonneville, A. & Kouzes, R. 2017. 3D cosmic ray muon tomography from an underground tunnel. *Pure and Applied Geophysics* 174(5): 2133–2141.
- Liu, J., Wang, Y., Feng, C., Liu, S. & Chen, Q. 2025. A back-end electronics based on optical fiber communication for small to medium-scale physics experiments. *Journal of Instrumentation* 20(05): T05002.
- Mao, X., Li, Z., Dong, S., Li, J., Zhang, J., Pang, J., Cheng, Y., Liao, B., Ouyang, X. & Han, R. 2023. Muon radiography experiments on the subway overburden structure detection. *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment* 1055: 168391.
- Procureur, S. 2018. Muon imaging: Principles, technologies and applications. *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment* 878: 169–179.
- Ronneberger, O., Fischer, P. & Brox, T. 2015. U-Net: Convolutional networks for biomedical image segmentation. In *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2015*. Vol. 9351. Lecture Notes in Computer Science. Cham: Springer, pp. 234–241.
- Siddon, R. L. 1985. Fast calculation of the exact radiological path for a three-dimensional CT array. *Medical Physics* 12(2): 252–255.
- Thompson, L. F., Stowell, J. P., Fargher, S. J., Steer, C. A., Loughney, K. L., O’Sullivan, E. M., Gluyas, J. G., Blaney, S. W. & Pidcock, R. J. 2020. Muon tomography for railway tunnel imaging. *Physical Review Research* 2(2): 023017.
- Wang, T., Wang, Y., Yao, Z., Liu, Y., Feng, C., Zhang, Z. & Liu, S. 2025. A compact readout electronics based on current amplifier for Micromegas detector in muon imaging. *IEEE Transactions on Nuclear Science* 72(3): 414–420.
- Zhu, C., Byrd, R. H., Lu, P. & Nocedal, J. 1997. Algorithm 778: L-BFGS-B: Fortran subroutines for large-scale bound-constrained optimization. *ACM Transactions on Mathematical Software* 23(4): 550–560.